Foundations of Predictive Modeling

Feature Engineering, Supervised Learning, Overfitting, Bias-Variance

Md Abdullah Al Kafi

Predictive Frameworks

Supervised learning involves learning a function mapping X (inputs) to y (labels) based on example pairs.

Regression

- Continuous outputs.
- e.g., Predicting stock prices, heights, or age.
- Goal: Minimize squared residuals.

Classification

- Discrete outputs.
- e.g., Image recognition, fraud detection.
- Goal: Maximize boundary separation.

Applying Feature Engineering

Raw data is rarely ready for modeling. Feature scaling ensures that all dimensions contribute equally to the distance calculation.

- **Normalization (Min-Max):**

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

  Rescales to a fixed range $[0, 1]$.

- **Standardization (Z-Score):**

$$X_{std} = \frac{X - \mu}{\sigma}$$

  Centers data around Mean 0 with Variance 1.

Application Rule: Use standardization for models that assume Gaussian distribution (e.g., Linear Regression, SVM).

Analyzing Model Fit

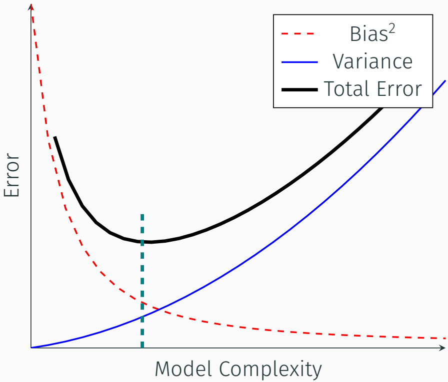
Model performance is evaluated by how well it generalizes to unseen data.

Metric	Underfitting	Overfitting
Training Error	High	Low
Test Error	High	High
Complexity	Too Simple	Too Complex
Root Cause	High Bias	High Variance

Analysis: A model that ”memorizes” the training set fails to capture the underlying pattern, leading to high variance.

The Tradeoff Curve

We seek the ”sweet spot” that minimizes Total Error.



The Fundamental Equation

The expected Prediction Error at a point x can be decomposed:

$$Err(x) = \mathbb{E}[Bias^2] + \mathbb{E}[Variance] + \sigma^2$$

- **Bias:** Error from erroneous assumptions in the learning algorithm.
- **Variance:** Error from sensitivity to small fluctuations in the training set.
- **Irreducible Error (σ^2):** Noise inherent in the problem itself.